A Model of Dutch

Part I
Game state, probability, and transitions

1 Cards and positions

Let

$$\mathcal{W} = \{w_1, \ldots, w_{52}\}, \qquad W = (w_1, \ldots, w_{52})^{\mathsf{T}}$$

be the set and catalogue vector of the 52 distinct cards. A card is the ordered pair $w_j = (\alpha_j, \beta_j)$, where $\alpha_j \in \{1, \ldots, 13\}$ is its rank (ace through king) and β_j is one of the four suits. Thus cards of the same rank but different suits are different elements of $\mathcal{W}$.

Fix the card axis once, for the entire model: list suits in the order $\clubsuit, \diamondsuit, \heartsuit, \spadesuit$, and within each suit list ranks $A, 2, \ldots, 10, J, Q, K$. If $r \in \{1, \ldots, 13\}$ is the rank index and $s \in \{1, \ldots, 4\}$ the suit index, then

$$w_{13(s-1)+r} = \text{the card of rank } r \text{ and suit } s.$$

For example, $w_7 = 7\clubsuit$, $w_{17} = 4\diamondsuit$, $w_{39} = K\heartsuit$, and $w_{40} = A\spadesuit$. Every card vector and the card columns of every matrix use this order. Let $S = (1, \ldots, 1) \in \mathbb{R}^{1\times 52}$.

Suppose there are $n \in \{2, \ldots, 10\}$ players. The collections are $\mathcal{C} = \{H_1, \ldots, H_n, D, P\}$: one hand for each player, the draw pile, and the discard pile. For each $c \in \mathcal{C}$, give collection c 52 possible positions,

$$\Lambda_c = \{(c, 1), (c, 2), \ldots, (c, 52)\}, \qquad \Lambda = \bigsqcup_{c\in\mathcal{C}} \Lambda_c, \qquad |\Lambda| = 52(n + 2).$$

For example, $(H_2, 3)$ is player 2's third hand position, and $(D, 1)$ is the top draw pile position at the initial deal.

Fix a round r, with $m = 0$ after dealing, the players' initial looks, and the starter discard reveal. Let $A_c^{(r,m)} = (a_{c,1}^{(r,m)}, \ldots, a_{c,52}^{(r,m)}) \in \{0, 1\}^{52}$ be the occupancy vector: $a_{c,k}^{(r,m)} = 1$ if position (c, k) holds a card and 0 if it is empty. The occupied positions are

$$L_c^{(r,m)} = \{(c, k) \in \Lambda_c : a_{c,k}^{(r,m)} = 1\}.$$

For a hand, $A_i^{(r,m)}$ abbreviates $A_{H_i}^{(r,m)}$. At the initial deal, each $A_{H_i}^{(r,0)}$ has ones in positions 1 through 4, $A_D^{(r,0)}$ has ones in positions 1 through $51 - 4n$, and $A_P^{(r,0)}$ has a one only in position 1. Thus

$$\sum_{c\in\mathcal{C}} |L_c^{(r,0)}| = 4n + (51 - 4n) + 1 = 52.$$

Adding a penalty card can occupy another hand position. Position labels stay fixed as cards move; an emptied position remains available for a later card.

2 Position probability vectors

For each possible position $\ell \in \Lambda$, let X_ℓ be its card when occupied and the symbol $\varnothing$ when empty. Given the observing player's information $\mathcal{I}$, define

$$V_\ell = (Q_{\ell 1}, \ldots, Q_{\ell,52}), \qquad Q_{\ell j} = \Pr(X_\ell = w_j \mid \mathcal{I}).$$

Let Q_c consist of the 52 rows $V_{(c,1)}, \ldots, V_{(c,52)}$. Grouping rows by collection gives

$$Q = \begin{pmatrix} Q_{H_1} \\ \vdots \\ Q_{H_n} \\ Q_D \\ Q_P \end{pmatrix} \in [0,1]^{52(n+2)\times 52}, \qquad Q_c \in [0,1]^{52\times 52}.$$

The 52 rows for each collection remain in Q even when some positions are empty. At the initial deal the occupancies are known, so a row sums to one if its position is occupied and to zero if empty:

$$\sum_{j=1}^{52} Q_{(c,k),j} = a_{c,k}^{(r,0)} \quad (c \in \mathcal{C},\ 1 \le k \le 52), \qquad \sum_{\ell \in \Lambda} Q_{\ell j} = 1 \quad (j = 1, \ldots, 52).$$

At the initial deal, exactly 52 rows are nonzero. In the setup sections below, Q abbreviates $Q^{(r,0)}$.

3 One player's initial information

Fix the observing player, indexed by o. Let $\ell_1, \ell_2 \in L_{H_o}^{(r,0)}$ be the two positions this player chooses to view at the start. Let $\ell_P = (P, 1)$ be the initial face up discard position. Suppose these positions contain three distinct observed cards a, b, d, respectively. The initial information is

$$\mathcal{I}_{r,0} = \{X_{\ell_1} = a,\ X_{\ell_2} = b,\ X_{\ell_P} = d\}.$$

For the initial model, assume a uniformly shuffled deck and a fixed dealing procedure.[1] Then the three observed rows of Q have a single entry equal to one. At each other occupied position, each of the 49 unobserved cards has probability $1/49$ and each observed card has probability zero. Rows for empty positions are zero.

For an example with three players, take the observing player to be player 1. Choose $\ell_1 = (H_1, 1)$ and $\ell_2 = (H_1, 2)$. Suppose the observer sees 7♣ at ℓ_1 and K♡ at ℓ_2, while the face up discard at ℓ_P is 4♢. Let e_j be the row vector with 52 entries with a one in the fixed card column j and zeros elsewhere. Then

$$V_{\ell_1} = e_7, \qquad V_{\ell_2} = e_{39}, \qquad V_{\ell_P} = e_{17}, \qquad V_\ell = \frac{S - e_7 - e_{39} - e_{17}}{49} \quad \text{at every other occupied position.}$$

At each empty position, $V_\ell = 0_{1\times 52}$.

[1] The shuffle model is provisional; a biased shuffle would change the initial probabilities.

Here is the full Q for this example in block form. Let $u = (S - e_7 - e_{39} - e_{17})/49$, and let $\mathbf{1}_q$ be a $q \times 1$ vector of ones:

$$Q_{H_1} = \begin{pmatrix} e_7 \\ e_{39} \\ u \\ u \\ 0_{48\times 52} \end{pmatrix}, \quad Q_{H_2} = Q_{H_3} = \begin{pmatrix} \mathbf{1}_4 u \\ 0_{48\times 52} \end{pmatrix},$$

$$Q_D = \begin{pmatrix} \mathbf{1}_{39} u \\ 0_{13\times 52} \end{pmatrix}, \qquad Q_P = \begin{pmatrix} e_{17} \\ 0_{51\times 52} \end{pmatrix}.$$

Stacking the five blocks gives the 260×52 matrix Q.

4 Collection probability vectors

For each collection $c \in \mathcal{C}$, define

$$K_{c,j} = \Pr(w_j \text{ is in collection } c \mid \mathcal{I}) = \sum_{\ell \in \Lambda_c} Q_{\ell j}, \qquad K_c = (K_{c,1}, \ldots, K_{c,52}).$$

Equivalently, $K_c = \sum_{\ell \in \Lambda_c} V_\ell$. Stacking these vectors gives the matrix of collection probabilities

$$K = \begin{pmatrix} K_{H_1} \\ \vdots \\ K_{H_n} \\ K_D \\ K_P \end{pmatrix} \in [0,1]^{(n+2)\times 52}.$$

At the initial deal, the entries of each K_c sum to its number of occupied positions:

$$\sum_{j=1}^{52} K_{c,j} = |L_c^{(r,0)}|.$$

For example, if w_j is one of the 49 unobserved cards, then

$$K_{H_o,j} = \frac{2}{49}, \qquad K_{H_i,j} = \frac{4}{49} \ (i \neq o), \qquad K_{D,j} = \frac{51 - 4n}{49}, \qquad K_{P,j} = 0.$$

These probabilities sum to one. Each of the two cards already seen in the observer's hand has $K_{H_o,j} = 1$, and the face up discard has $K_{P,j} = 1$.

For the example with three players, write $U = S - e_7 - e_{39} - e_{17}$. The five collection rows are

$$\begin{aligned} K_{H_1} &= e_7 + e_{39} + \tfrac{2}{49} U, \\ K_{H_2} &= \tfrac{4}{49} U, \qquad K_{H_3} = \tfrac{4}{49} U, \\ K_D &= \tfrac{39}{49} U, \qquad K_P = e_{17}. \end{aligned}$$

Conservation of each card gives

$$\sum_{c \in \mathcal{C}} K_c = S. \tag{1}$$

The jth entry says that card w_j is somewhere in the game with probability one.

5 Round scores

Let $v \in \mathbb{R}^{52}$ be the fixed card value column vector, with

$$v_j = \begin{cases} 0, & w_j \text{ is a red king,} \\ \alpha_j, & \text{otherwise.} \end{cases}$$

Thus an ace is worth 1, a jack 11, a queen 12, a black king 13, and a red king 0. At event (r, m), define $K_c^{(r,m)} = \sum_{\ell \in \Lambda_c} Q_{\ell,:}^{(r,m)}$. The observer's expected current hand points for player i are

$$\mathbb{E}[\text{hand points of } i \mid \mathcal{I}_{r,m}] = K_{H_i}^{(r,m)} v.$$

At the end of round r, all remaining hand cards are revealed. The round score and cumulative score are then

$$s_{i,r} = K_{H_i}^{(r,\text{end})} v, \qquad C_{i,r} = \sum_{q=1}^{r} s_{i,q}.$$

After the reveal, the hand rows of Q are exact card indicators, so $s_{i,r}$ is the actual score rather than an expectation. Writing $\mathbf{s}_r = (s_{1,r}, \ldots, s_{n,r})^\mathsf{T}$ and $\mathbf{C}_r = (C_{1,r}, \ldots, C_{n,r})^\mathsf{T}$ gives the update across rounds

$$\mathbf{C}_r = \mathbf{C}_{r-1} + \mathbf{s}_r, \qquad \mathbf{C}_0 = \mathbf{0}_n.$$

If $\max_i C_{i,r} < 100$, a new round resets the card locations and beliefs to the initial setup while keeping $\mathbf{C}_r$. If $\max_i C_{i,r} \geq 100$, determine the winner from the lowest cumulative score, using the Dutch caller's tie rule when the caller is among the players tied for lowest. For any remaining tie, compare only the tied players' scores in the most recent earlier round ended by Dutch. Favor the lowest score in that round; if no such round exists or the tie remains, play another round even though the 100 point threshold was reached.[2]

6 Dependence between positions

The matrix Q records individual position probabilities, not a claim that positions are independent. For example, if ℓ and ℓ' are two different unobserved positions and u and v are two different unobserved cards, uniform shuffling gives

$$\Pr(X_\ell = u,\ X_{\ell'} = v \mid \mathcal{I}_{r,0}) = \frac{1}{49 \cdot 48}, \qquad \Pr(X_\ell = u,\ X_{\ell'} = u \mid \mathcal{I}_{r,0}) = 0.$$

Multiplying the two marginal entries $1/49$ would give neither probability correctly. The uniform shuffle gives this relationship because cards are dealt without replacement. In general, Q alone does not determine joint probabilities; the update rule below uses the law of sampling without replacement created by the initial shuffle and preserved by visible row movements when player choices are not used as evidence.

7 Turn state and event history

Within each round r, set $m = 0$ after the initial setup and increase m after each subsequent card movement, observation, or control event. Thus $Q^{(r,m)}$ is the full position by card matrix after

[2] The meaning of "previous Dutch round" and the extra round after an unresolved tie are provisional house rules.

event m of round r; m resets at the next round. Its dimensions remain $52(n+2) \times 52$; rows for newly empty positions become zero, and newly occupied rows become nonzero. Let $\mathcal{I}_{r,m}$ be the information available to observer o after that event. When occupancy is known, it can be read from row sums:

$$a_{c,k}^{(r,m)} = \sum_{j=1}^{52} Q_{(c,k),j}^{(r,m)}.$$

To select whose turn is in progress, let $f_i \in \mathbb{R}^n$ be the ith player basis vector and define

$$T^{(r,m)} = f_i f_i^{\mathsf{T}} \in \{0,1\}^{n \times n} \quad \text{when it is player } i\text{'s turn.}$$

Exactly one diagonal entry of $T^{(r,m)}$ is one. A matching discard can be performed by another player without changing whose turn it is, so the acting player is recorded separately for each event.

Let $\phi_{r,m}$ identify the phase of the turn, and let $\rho_{r,m}$ be the position of a drawn card being held temporarily, or $\varnothing$ when no card is held. Let $b_{r,m} \in \{0,1\}^n$ record eligibility to attempt another matching discard during the current turn: entry $b_{r,m,i} = 1$ means player i may attempt a match while a matching window is open; entry 0 means that player is barred. At the start of each turn $b_{r,m} = \mathbf{1}_n$. A failed attempt sets the acting player's entry to zero. Let $\delta_{r,m} \in \{0, 1, \ldots, n\}$ identify the player who placed the current top discard; $\delta_{r,0} = 0$ for the starter card, which no player discarded. Let $\gamma_{r,m} \in \{0, 1, \ldots, n\}$ be the Dutch caller (0 means no call), and let $\nu_{r,m} \in \{0, 1, \ldots, n\}$ count final turns still to be completed, including the turn currently in progress. Initially $\gamma_{r,0} = \nu_{r,0} = 0$. The cumulative score vector $\mathbf{C}_{r-1}$ is fixed during round r and updated when that round is scored. A state snapshot is

$$\mathcal{X}_{r,m} = (Q^{(r,m)}, T^{(r,m)}, \phi_{r,m}, \rho_{r,m}, b_{r,m}, \delta_{r,m}, \gamma_{r,m}, \nu_{r,m}, \mathbf{C}_{r-1}).$$

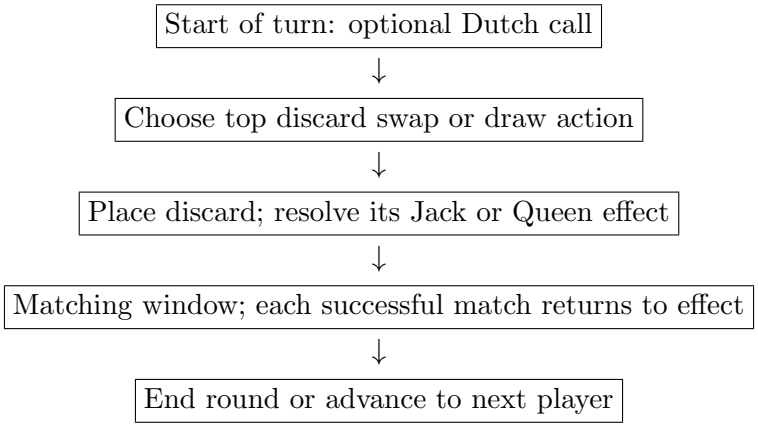
Let $\epsilon_{r,m}$ record the event type, acting player, source and destination positions, and any card identity observed. The history

$$\mathcal{H}_{r,m} = (\mathcal{X}_{r,0}, \epsilon_{r,1}, \mathcal{X}_{r,1}, \ldots, \epsilon_{r,m}, \mathcal{X}_{r,m})$$

retains both the time series of probability matrices and the actions that produced it. This will allow a later behavior model to use earlier choices and observations.

7.1 Turn control operators

Use the phase labels $\phi \in \{\text{start}, \text{action}, \text{held}, \text{effect}, \text{matching}, \text{turn end}, \text{round end}\}$. The action section below specifies the card movement and observation events. One turn follows:[3]



[3]The order of Jack and Queen effects before matching responses is provisional.

Let $\Pi \in \{0,1\}^{n\times n}$ be the clockwise permutation,

$$\Pi f_i = f_{i+1}\ (i<n), \qquad \Pi f_n = f_1.$$

At the start of a turn, reset $b = \mathbf{1}_n$. If the round continues after that turn, advance the selector by

$$T' = \Pi T \Pi^{\mathsf{T}}, \qquad \phi' = \text{start}.$$

The starting player of each round is supplied as an initial selector $T^{(r,0)}$.

At the start of their turn, player i may call Dutch if no call is active. Set

$$(\gamma', \nu') = (i, n).$$

The caller's current turn is the first of the n final turns. After each completed turn, decrease ν by one; when it reaches zero, end the round. During these turns, another player's Jack cannot swap a card in the caller's hand, but the caller may still match another player's discard.

A player j may attempt a match when $\phi = \text{matching}$, the draw pile is nonempty, $b_j = 1$, and $j \neq \delta$. Each new discard by player i sets $\delta' = i$. A failed attempt sets $b'_j = 0$; a successful match leaves b unchanged. An incorrect attempt leaves δ unchanged, so other eligible players may still respond. If the draw pile empties, set $b' = \mathbf{0}_n$, finish the current action and any card effect, then end the round without advancing T.[4]

8 Actions and information updates

Each arrow below is a separate event and produces a new $Q^{(r,m)}$. All probabilities are from the viewpoint of the fixed observer o.

8.1 Movement and observation operators

Let $M = 52(n+2)$. For every possible position $x \in \Lambda$, let $g_x \in \mathbb{R}^M$ be the basis vector with a one in the coordinate for x and zeros elsewhere. In particular, g_a and g_b refer to the vectors for positions a and b. For distinct positions $a, b \in \Lambda$, exchanging their rows uses the permutation matrix

$$R_{a,b} = I_M - (g_a - g_b)(g_a - g_b)^{\mathsf{T}}, \qquad \mathcal{M}_{a,b}(Q) = R_{a,b}Q.$$

If one row is empty, $\mathcal{M}_{a,b}$ moves a card into it; if both are occupied, it swaps the cards' position distributions.

When the observer sees card w_j at position ℓ, apply $\mathcal{O}_{\ell,j}$. The general rule, after any movement, is Bayes' rule:

$$Q^{+}_{\ell j} = \frac{\Pr(\widetilde{X}_\ell = w_j,\ O \mid \mathcal{I}_{r,m})}{\Pr(O \mid \mathcal{I}_{r,m})}, \qquad \Pr(O \mid \mathcal{I}_{r,m}) > 0,$$

where O is the new observation and $\widetilde{X}$ is the card layout after movement. A movement with no new card information uses $\mathcal{M}$ alone. Another player's private look, a turn change, or a Dutch call does not directly change the observer's Q.

In the current baseline, cards start from a uniform shuffle, players cannot voluntarily rearrange them, and Jack swaps show the positions exchanged. We record player choices but do not use them as evidence about hidden cards. Consequently, the unknown cards remain uniformly distributed

[4] Stopping matching as soon as the draw pile empties is a house rule.

over the unknown occupied positions. Let $\mathcal{F}$ be known pairs (ℓ, j) of positions and cards, $\mathcal{U}$ the unknown occupied positions, and $\mathcal{J}$ the unseen card indices. With $N = |\mathcal{U}| = |\mathcal{J}|$ and

$$g_{\mathcal{U}} = \sum_{\ell \in \mathcal{U}} g_\ell, \qquad h_{\mathcal{J}} = \sum_{j \in \mathcal{J}} e_j,$$

the full matrix is, for $N > 0$,

$$Q = \sum_{(\ell,j) \in \mathcal{F}} g_\ell e_j + \frac{1}{N} g_{\mathcal{U}} h_{\mathcal{J}}. \tag{2}$$

Empty positions have zero rows; when $N = 0$, the second term vanishes. A visible movement permutes the row labels in $\mathcal{F}$ and $\mathcal{U}$ without changing $\mathcal{J}$.

If an unknown position $\ell \in \mathcal{U}$ is revealed as w_j, $j \in \mathcal{J}$, remove ℓ from $\mathcal{U}$ and j from $\mathcal{J}$, and add (ℓ, j) to $\mathcal{F}$. For $N > 1$, the observation operator is

$$\mathcal{O}_{\ell,j}(Q) = Q - \frac{1}{N} g_{\mathcal{U}} h_{\mathcal{J}} + g_\ell e_j + \frac{1}{N-1}(g_{\mathcal{U}} - g_\ell)(h_{\mathcal{J}} - e_j). \tag{3}$$

For $N = 1$, replace the unknown block by $g_\ell e_j$. If w_j was already known at ℓ, $\mathcal{O}_{\ell,j}$ leaves Q unchanged. A new reveal makes that row certain and leaves each of the other $N-1$ unknown rows uniform over the remaining $N-1$ cards. This rule is exact under the baseline assumptions above; arbitrary behavioral evidence may require more than Q.

8.2 Event rules

Let d be the lowest occupied draw pile position (the top), p the highest occupied discard position (the top), p^+ the next empty discard position, h a chosen occupied hand position, and u the next empty hand position of the acting player. Read each operator sequence from left to right in time. When a face up card is shown, its exact rank and suit are observed.

Round setup.
A new shuffle resets the prior. The observer's two initial looks use $\mathcal{O}_{\ell,j}$; the initial face up discard uses $\mathcal{O}_{(P,1),j}$. Other players' private initial looks leave the observer's Q unchanged. The starter card triggers no Jack or Queen effect. These observations define $Q^{(r,0)}$.

Take the top discard.
If the top discard w_j at p is exchanged with hand position h, and the replaced card is revealed as w_k, apply $\mathcal{M}_{p,h} \to \mathcal{O}_{p,k}$. Afterwards $Q_{h,:} = e_j$ and $Q_{p,:} = e_k$. The player who made this discard cannot immediately match against it, even if the card just taken into h has the same rank.

Draw.
Apply $\mathcal{M}_{d,u}$. The drawer sees the card; apply $\mathcal{O}_{u,j}$ only if the drawer is o. Set $\rho = u$ while the card occupies that temporary hand position.

Discard the draw.
Apply $\mathcal{M}_{u,p^+} \to \mathcal{O}_{p^+,j}$. Clear ρ. The face up reveal may be redundant for o if o drew it.

Swap the draw.
After drawing, apply $\mathcal{M}_{u,h} \to \mathcal{M}_{u,p^+} \to \mathcal{O}_{p^+,k}$. The drawn card ends at h; the replaced hand card w_k is revealed on the discard pile. Clear ρ.

Successful match.
While the draw pile is nonempty and the acting player differs from δ, apply $\mathcal{M}_{h,p^+} \to \mathcal{O}_{p^+,j}$. Set δ to the acting player. Each new discard may open a response from another player. Responses are logged in their observed order; b is unchanged.

Incorrect match.
Reveal the offered card at its unchanged hand position, then draw an unseen penalty card:[5] $\mathcal{O}_{h,j} \to \mathcal{M}_{d,u}$. The reveal is public, so $\mathcal{O}_{h,j}$ applies even when an opponent made the attempt: row h becomes e_j, and the other unknown rows exclude w_j. Set the acting player's matching eligibility to zero for this turn. The offered card returns face down to h; its known identity is retained and it triggers no special effect.

Unauthorized look.
For each card viewed, apply $\mathcal{O}_{h,j}$ only if o is the viewer, then $\mathcal{M}_{d,u}$ for its unseen penalty card. Another player's private look is logged without revealing that card to o.

Jack and Queen.
A discarded Jack applies $\mathcal{M}_{a,b}$ to two visibly selected face down positions. A discarded Queen applies $\mathcal{O}_{h,j}$ only if its user is o; otherwise Q does not change. These effects also apply to successful matching discards and resolve before matching responses.

Control and round end.
Turn changes, Dutch calls, and matching eligibility do not change Q in this model. At round end, apply $\mathcal{O}_{h,j}$ to each remaining unknown hand card and compute the scores.

Player choices and decisions not to act may eventually be used as evidence: which source or hand position was chosen, whether a match was attempted, Jack and Queen targets, and whether Dutch was called. That behavioral inference is deferred; each choice is still recorded in the event history.

Part II
Decisions

9 Immediate expected value of hand points

Fix a turn of the observing player o, and let $H = L_{H_o}^{(r,m)}$ be the occupied hand positions before acting. For any $h \in H$, the estimated points in that position are

$$\mu_h(Q) = Q_{h,:}v = \sum_{j=1}^{52} Q_{h,j}v_j.$$

If the card is known to be w_k, then $\mu_h(Q) = v_k$; otherwise this is its value weighted by probability. The value of an action in this first decision model is the expected reduction in the observing player's hand points immediately after the action. A larger value is better.

[5] We assume that penalty cards come from the draw pile and are not viewed by their recipients.

9.1 Take the top discard

Suppose the visible top discard at position p is w_k. Taking it requires swapping it with some $h \in H$. If h actually holds w_j, the point reduction is $G_P(h, j) = v_j - v_k$. Before seeing that hand card, its expected reduction and the best swap are

$$\mathrm{EV}_P(h) = \sum_{j=1}^{52} Q_{h,j} G_P(h, j) = \sum_{j=1}^{52} Q_{h,j}(v_j - v_k) = \mu_h(Q) - v_k,$$

$$\mathrm{EV}_P^* = \max_{h \in H} \mathrm{EV}_P(h) = \max_{h \in H} \sum_{j=1}^{52} Q_{h,j} G_P(h, j).$$

When the card at h is unknown, $Q_{h,j}$ weights the point reduction for each card it might be. When it is known to be w_ℓ, $Q_{h,\ell} = 1$ and the same formula gives $\mathrm{EV}_P(h) = v_\ell - v_k$. Thus the best target is a hand position with the largest estimated value. A negative EV_P^* means even that swap is expected to increase hand points; it does not make the swap optional after the player chooses to take the discard. This action is unavailable if H is empty.

9.2 Draw, then discard or swap

Let d be the top draw position and u the temporary empty hand position used for the draw. Before drawing, card w_j has probability $Q_{d,j}$. If the observing player draws and sees w_j, update the belief matrix first:

$$Q^{[j]} = \mathcal{O}_{u,j}(\mathcal{M}_{d,u}(Q)) \qquad (Q_{d,j} > 0).$$

The observation can change the estimates of cards still unknown in H. After seeing w_j, discarding it leaves the hand points unchanged. Swapping it with $h \in H$ has conditional expected reduction $\mu_h(Q^{[j]}) - v_j$. The best response to this draw is

$$G_D(j) = \max\left\{0,\ \max_{h \in H}\left(\mu_h(Q^{[j]}) - v_j\right)\right\}.$$

Choose a maximizing swap when its reduction is positive; otherwise discard the drawn card. If H is empty, define $G_D(j) = 0$. Before seeing the draw, the expected value of choosing to draw is

$$\mathrm{EV}_D(Q) = \sum_{j : Q_{d,j} > 0} Q_{d,j} G_D(j).$$

The branch that discards the draw immediately has zero immediate reduction in hand points, but the draw option may have positive value because the player can decide whether to swap after seeing the card.

When both turn options are legal, this policy takes the top discard if $\mathrm{EV}_P^* > \mathrm{EV}_D(Q)$ and otherwise draws. For example, if the top discard is worth 4 points, replacing a known card worth 9 points gives $\mathrm{EV}_P(h) = 5$, while replacing an unknown card estimated at 7 points gives $\mathrm{EV}_P(h) = 3$.

This first policy counts only the observing player's immediate hand points. Matching opportunities, Jack and Queen effects, round and game outcomes, and information gained for later decisions are not yet included in its action value. A later section can add a value for reducing uncertainty about hidden card locations; its information measure and weight remain to be chosen. Lots of further work using information theory ideas can be used to opitmize for better Dutch outcomes.

Part III
Behavioral updates

Coming soon.